Autonomous Retail Researcher Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Problem Statement & Objectives

1. Problem Statement

The retail industry is highly dynamic and data-driven, involving continuous monitoring of consumer trends, pricing strategies, competitor analysis, product demand, and market behavior. Retail research traditionally requires manual collection of data from multiple sources such as e-commerce platforms, market reports, customer reviews, and industry news. This process is time-consuming, inconsistent, and often leads to fragmented insights.

With the exponential growth of digital retail data, there is a need for an intelligent system that can autonomously gather, analyze, and summarize retail-related information. The absence of automation results in delayed insights, inefficient decision-making, and missed business opportunities.

To address this challenge, this project proposes the development of an LLM-powered Autonomous Retail Researcher Agent using LangChain and CrewAI. The system is capable of independently processing user queries, navigating the web, synthesizing multi-source information, and generating structured insights stored in a knowledge repository.

2. Project Objectives

· To design and develop an intelligent autonomous agent using Agentic AI principles

· To automate the process of retail market research and data collection

· To enable the agent to understand user queries and decompose them into actionable tasks

· To integrate web search tools for real-time retail data retrieval

· To synthesize data from multiple high-quality sources

· To generate structured, summarized insights for decision-making

· To store research outputs in a persistent knowledge repository for future use

· To improve efficiency in retail analytics and business intelligence

3. **Scope of the Project**

The scope of this project includes:

· Automation of retail research processes

· Web-based information retrieval across e-commerce and market sources

· Multi-source data aggregation and intelligent summarization

· Knowledge storage and retrieval for historical insights

· Natural language-based user interaction

Out of Scope:

· Direct integration with live e-commerce transaction systems

· Real-time inventory or supply chain execution

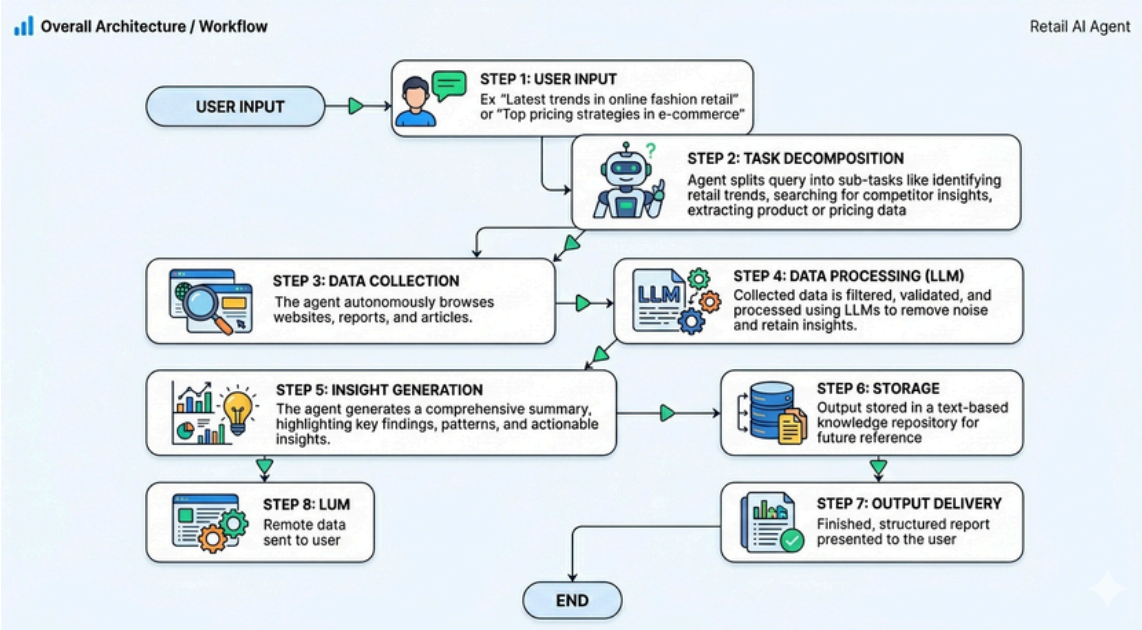
· Advanced predictive analytics using real-time sales data

Proposed Solution

1. Key features

·Autonomous query understanding and task planning
·Multi-agent collaboration using CrewAI
·Web navigation and search integration for real-time retail insights
·Intelligent summarization using Large Language Models (LLMs)
·Knowledge repository for storing structured outputs
·Automated report generation with key findings
· Scalable and modular architecture

2. OverallArchitecture / Workflow



Step 1: User Input

The user provides a retail-related query (e.g., "Latest trends in online fashion retail" or "Top pricing strategies in e-commerce").

Step 2: Task Decomposition

The agent breaks the query into smaller sub-tasks such as:

- Identifying relevant retail trends
- Searching for competitor insights
- Extracting product or pricing data

Step 3: Data Collection

Using integrated web search tools, the agent autonomously browses and collects data from multiple retail sources such as websites, reports, and articles.

Step 4: Data Processing

The collected data is filtered, validated, and processed using LLMs to remove noise and retain relevant insights.

Step 5: Insight Generation

The agent generates a comprehensive summary highlighting key findings, patterns, and actionable insights.

Step 6: Storage

The generated output is stored in a text-based knowledge repository for future reference.

Step 7: Output Delivery

The final structured report is presented to the user in a readable and organized format.

3. **Tools & Technologies Used**

- Python : Core programming language
- LangChain : Framework for building LLM-powered applications
- CrewAI : Multi-agent orchestration and collaboration
- LLM APIs : Natural language understanding and generation
- Web Search Tools : Real-time data retrieval from retail sources
- Text Storage Systems : Knowledge persistence and retrieval

Results & Output

1. Screenshots / outputs

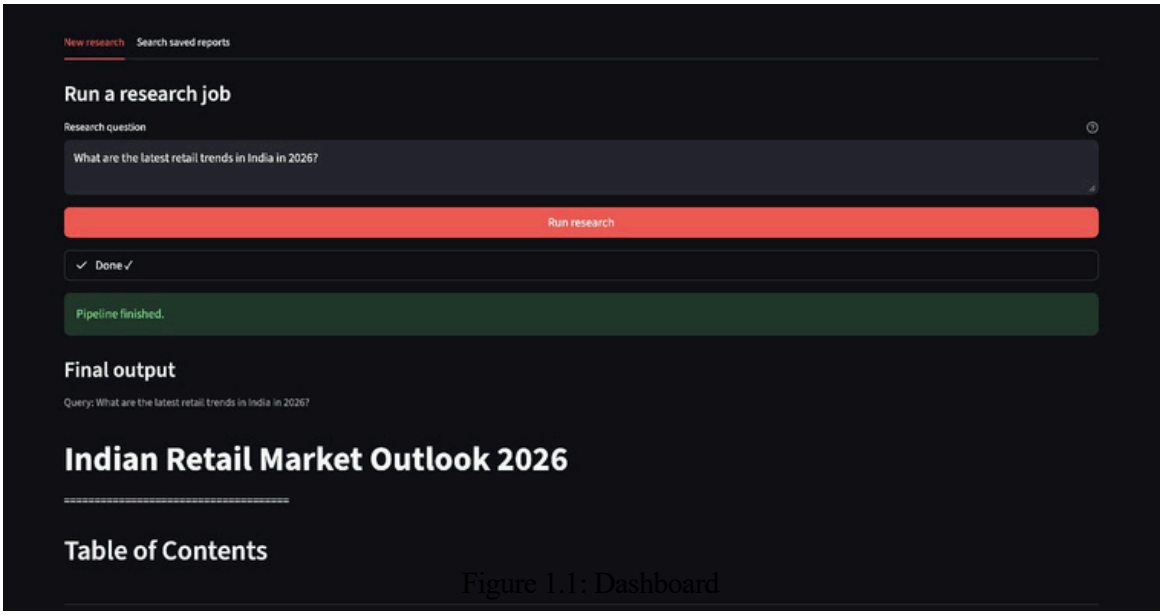


Figure 1.1: Dashboard

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Retail Research Agent ☆ Start Page

Deploy

Summary

The Indian retail market is expected to experience significant growth, reaching USD 1.4 trillion by 2026, with a projected CAGR of 12.2%. This growth is driven by rapid urbanization, rising e-commerce adoption, and the increasing influence of social media on consumer shopping habits. Social commerce and influencer marketing are expected to revolutionize the Indian retail landscape, providing new channels for businesses to reach consumers.

Key Insights

- Rapid urbanization and increasing e-commerce adoption are expected to drive the growth of the Indian retail market.
- Social commerce and influencer marketing could provide new channels for businesses to reach consumers.
- E-commerce and social media could provide opportunities for businesses to expand their customer base.

Market Trends

Indian Retail Market Growth

- The Indian retail market is expected to reach USD 1.4 trillion by 2026 (Source: https://www.rhbsec.com/images/2026-02-22_India-Retail-Market-report.pdf).
- The market is projected to grow at a CAGR of 12.2% (Source: https://www.rhbsec.com/images/2026-02-22_India-Retail-Market-report.pdf).
- Alternatively, Euromonitor forecasts the Indian retail market to grow by 13.1% in 2026 (Source: https://www.euromonitor.com/india-retail-market-outlook/report).

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Deploy

Risks

- The high growth expectations for the Indian retail market may indicate potential risks, such as over-expansion or competition.
- The increasing adoption of e-commerce could lead to concerns about job displacement or changes in consumer behavior.

Opportunities

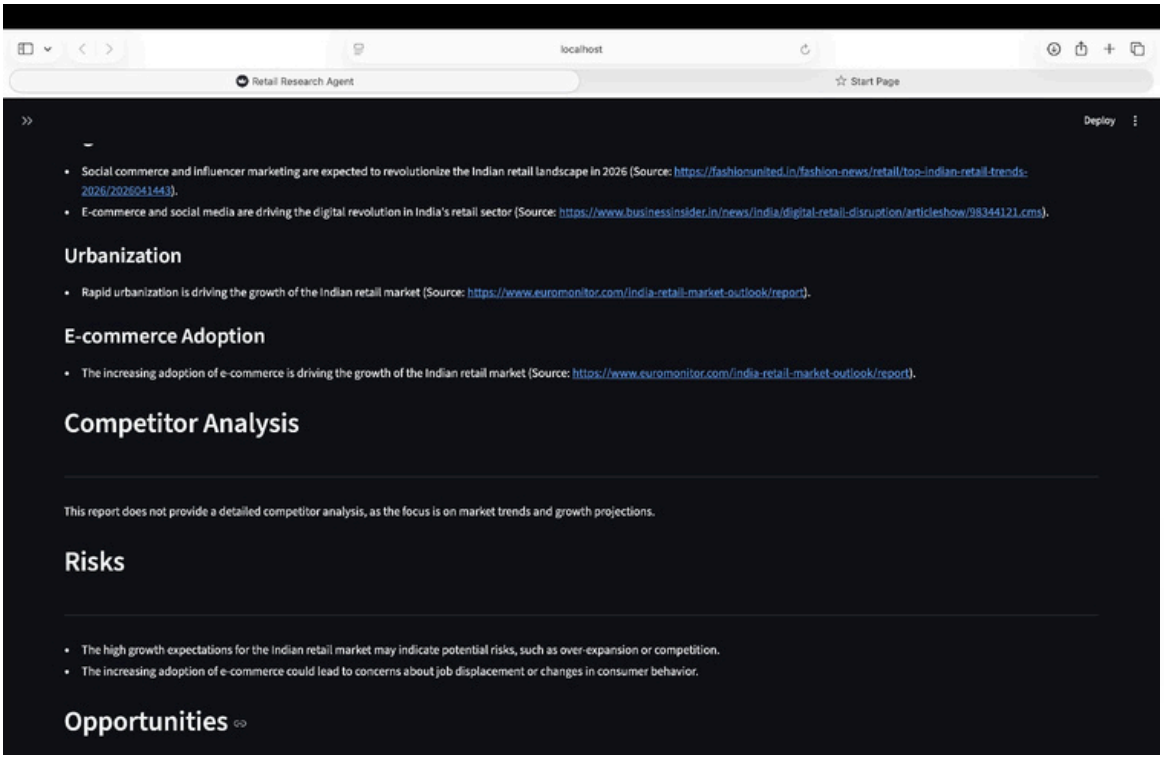
- Social commerce and influencer marketing could provide new channels for businesses to reach consumers.
- E-commerce and social media could provide opportunities for businesses to expand their customer base.

Sources

- https://www.rhbsec.com/images/2026-02-22_India-Retail-Market-report.pdf - Indian retail market size to reach USD 1.4 trillion by 2026
- https://www.euromonitor.com/india-retail-market-outlook/report - Indian retail market to grow by 13.1% in 2026
- https://fashionunited.in/fashion-news/retail/top-indian-retail-trends-2026/2026041443 - Social commerce and influencer marketing to revolutionize Indian retail landscape in 2026
- https://www.businessinsider.in/news/india/digital-retail-disruption/articleshow/98344121.cms - E-commerce and social media driving digital retail disruption in India

2. Reports / dashboards / models

- Generated retail research summaries in structured text format
- Insights categorized with headings such as trends, competitors, and pricing strategies
- Persistent knowledge repository storing past research outputs



3. Keyoutcomes

- Significant reduction in manual retail research effort
- Faster and more efficient information retrieval
- Improved accuracy through AI-based summarization
- Scalable solution adaptable to multiple retail domains
- Practical implementation of Agentic AI in real-world retail scenarios

Conclusion

The Autonomous Retail Researcher Agent demonstrates the effectiveness of Agentic AI in transforming retail research processes. By leveraging LLMs along with LangChain and CrewAI, the system autonomously processes user queries, gathers relevant data from multiple sources, and generates meaningful insights.
This project highlights how AI can enhance efficiency, reduce manual workload, and support better decision-making in the retail industry. It also establishes a strong foundation for deploying autonomous agents in real-world business applications.

Future Scope & Enhancements

- Integration with real-time e-commerce APIs
- Advanced predictive analytics for demand forecasting
- Voice-based interaction with the agent
- Interactive dashboards for visualization
- Multi-domain research expansion beyond retail
- Integration with enterprise retail systems (ERP, CRM)
- Enhanced memory systems for long-term learning and personalization